

COSIMA Cookbook: a community-driven resource for ocean and sea-ice modelling science

Navid C. Constantinou^{1,14*}, Julia Neme^{2*}, Wilton Aguiar², Matthis Auger¹³, Ashley J. Barnes^{4,10}, Romain Beucher³, Dhruv Bhagtani¹¹, Christopher Yit Sen Bull³, Hannah Dawson¹⁷, Noah Day¹⁴, Fabio Boeira Dias¹⁵, Edward W. Doddridge¹⁷, Anupiya Ellepola², Denisse Fierro-Arcos¹⁷, Angus Gibson², Maurice F. Huguenin⁸, Aidan Heerdegen³, Andy McC. Hogg^{2,3,4}, Ryan M. Holmes⁵, Wilma Huneke^{2,10}, Jemma Jeffree^{2,4}, Andrew E. Kiss², Minghang Li³, Josué Martínez-Moreno¹², Jan Jaap Meijer¹⁷, Thomas Moore⁷, Ruth Moorman⁶, Paige Martin³, Adele K. Morrison², James Munroe¹, Aditya Narayanan¹⁶, Micael Oliveira³, Ellie Q. Y. Ong^{4,10}, Madelaine G. Rosevear^{4,17}, Christina Schmidt¹², Taimoor Sohail¹⁴, Paul Spence^{4,17}, Dougal T. Squire³, Anton Steketeer³, Charles Turner³, Felipe Vilela da Silva⁹, Marc White^{2,3}, Luwei Yang², Claire Yung², and Jan Zika¹⁵

¹ Earthdata Associates, Canada ² Australian National University, Australia ³ Australia's Climate Simulator (ACCESS-NRI), Australian National University, Canberra, Australia ⁴ Australian Research Council Centre of Excellence for the Weather of the 21st Century, Australia ⁵ Bureau of Meteorology, Australia ⁶ California Institute of Technology, USA ⁷ Commonwealth Scientific and Industrial Research Organisation, Australia ⁸ GEOMAR Helmholtz Centre for Ocean Research Kiel, Germany ⁹ IFREMER - French Institute for Ocean Science, France ¹⁰ Monash University, Australia ¹¹ Princeton University, USA ¹² British Antarctic Survey, UK ¹³ University of Brest, France ¹⁴ University of Melbourne, Australia ¹⁵ University of New South Wales Sydney, Australia ¹⁶ University of Southampton, UK ¹⁷ University of Tasmania, Australia * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#))

Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks (Kluyver et al., 2016). It has been developed by the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA; <https://cosima.org.au>) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean–sea ice model configurations (e.g. Kiss et al., 2020). It is developed openly in a GitHub repository and it combines introductory tutorials with worked analysis examples as Python Jupyter notebooks. All tutorials and examples are exposed through a browsable documentation site built with Sphinx, a documentation generator for Python projects (Komiya et al., 2025) (Figure 1). Tagged versions of the repository are uploaded to a citable archived release (Constantinou et al., 2026).

COSIMA Cookbook

Search Go

Navigation

Lessons

Cooking Tutorials

- Basics
- Advanced

Recipes

Easy Recipes

Advanced Recipes

Regional Specialties

Resources

Contributing

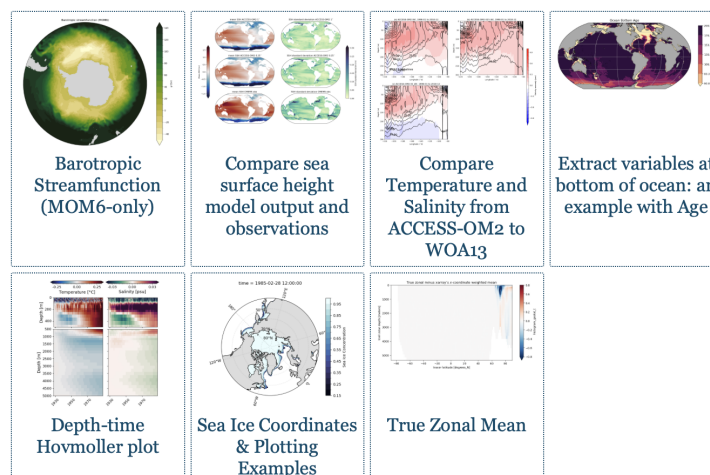
- Getting started with git
- Working on your recipe
- Tips for making a good recipe
- Submitting a Pull Request
- Reviewing existing Pull Requests

Notebook Guidelines

- Thumbnails
- Images in notebooks

GitHub Repository

Easy Recipes



latest

©2025, COSIMA. | Powered by Sphinx 9.1.0 & Alabaster 1.0.0 | [Page source](#)

Figure 1: The COSIMA Cookbook documentation website, showing the browsable Sphinx-based landing page used to navigate tutorials and recipes. The documentation website lives at <https://cosima-recipes.readthedocs.io>.

The COSIMA Cookbook is deliberately organized using a gastronomy theme to help users quickly understand and navigate the website. “Cooking Tutorials” refers to tutorials that teach generic, transferable techniques for handling ocean–sea ice model output. Then comes the main part of any cookbook – its recipes. Here, by “recipes” we mean self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. First come the “Easy Recipes” that provide a good starting point after the tutorials; then the “Advanced Recipes” that are more elaborate and demonstrate advanced analysis techniques, and lastly the “Regional Specialties” contains recipes for regional model configurations (Barnes et al., 2024). Together these sections present the documentation as a browsable collection of lessons and recipes gathered into a single cookbook. The COSIMA Cookbook is one of several examples of using a gastronomy theme for computational workflows, for example Project Pythia Cookbooks for the general geoscience community and the FAIR Cookbook for working with life science data.

Pedagogy and instructional structure are central to the project. The recipes provide the starting point for more elaborate analyses. “Cooking Tutorials” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused recipes giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

Within this structure, the tutorials leverage and demonstrate open source tools for scientific analysis of large data. This includes examples of loading model output through Intake-based

catalogues (Intake developers, 2026), using Xarray (Hoyer & Hamman, 2017) and Dask (Dask Development Team, 2016; Rocklin, 2015) to process very large multi-dimensional datasets in parallel. The recipe collection includes a Lagrangian particle-tracking workflow built with Parcels (Delandmeter & Seville, 2019), as well as analyses that use xgcm (Abernathy et al., 2022) to handle variables defined on staggered finite-volume grids, and xESMF (Zhuang et al., 2025) for regridding model output. Figures are generated using matplotlib (Hunter, 2007) incorporating Cartopy (Elson et al., 2024) for map-based visualisation.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and reproducible end-to-end workflows remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook addresses this gap by packaging reusable workflows in the same medium in which users actually work.

Statement of Need

Ocean and climate model analysis has a steep entry cost. New users must learn how to find datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and produce scientifically meaningful diagnostics. General-purpose libraries such as Xarray (Hoyer & Hamman, 2017) provide the foundations for these tasks, but they do not by themselves show a learner how to translate a research question into a robust analysis workflow for a specific model and computing environment.

That gap is not only pedagogical; it directly affects how quickly model output can be turned into science. For a beginner, becoming productive with a model often requires substantial time just to understand the structure of the output, the relevant conventions, and the computational environment in which analysis is expected to run. There is also substantial scientific labour embedded in many common diagnostics. Some calculations, for example computing cross-contour transports, require non-trivial development, understanding of model numerics, and validation before they can be used confidently. If every new user or project had to reconstruct those workflows independently, a large amount of effort would be repeatedly spent on rebuilding analysis code rather than conducting research using the model output.

The COSIMA Cookbook facilitates knowledge sharing and accelerates research with a domain-specific, openly maintained collection of computational lessons and examples. Its contribution is not a new analysis library; rather, it is a reusable scientific and educational resource that exposes complete, well-documented workflows in the same notebook format in which many researchers actually explore data and communicate methods. By choosing notebooks instead of packaging these examples only as a software library, the project makes intermediate reasoning, methodological choices, and practical execution details visible to learners while still giving more experienced users analysis patterns they can adapt directly for research. This aligns well with JOSE's emphasis on open educational resources that enable computational learning through authentic practice.

Several design decisions make the Cookbook particularly useful for adoption by other groups. First, each notebook is intended to be self-contained and well-documented, so learners can read, run, and modify a complete workflow without reconstructing missing context from scattered notes. Second, the repository distinguishes tutorials from recipes. Tutorials teach transferable skills, while recipes demonstrate how those skills combine in realistic analysis tasks. Third, the project includes contributor guidance and notebook review conventions that encourage new analyses to be written in a reusable and pedagogically clear style. A caveat is that some recipes use high-resolution model output with datasets

108 of order terabytes, so the underlying data are not always practical to distribute or mirror
109 in full. However, beyond the initial data-loading step, which usually occurs early in each
110 recipe, the analysis workflows are generally applicable to most operational ocean-sea ice
111 model outputs that can be loaded with xarray (Hoyer & Hamman, 2017).

112 Thus, the structure is valuable beyond the immediate COSIMA context. Many research
113 communities maintain model output on shared infrastructure and face the same challenge:
114 turning expert tacit knowledge into examples that newcomers can adapt. The Cookbook
115 offers a model for how a scientific collaboration can capture that knowledge in version-
116 controlled notebooks, publish it as a living resource, and continuously improve it through
117 community contribution.

118 Educational Design and Experience of Use

119 The learning experience is organised as a progression. The documentation points new users
120 first to introductory lessons, including material on loading, slicing, and visualising model
121 output. Learners then move to more advanced tutorials and finally to recipe notebooks
122 that address concrete scientific questions. The categories of “easy” and “advanced” recipes
123 provide a lightweight pedagogical cue about expected complexity and scope, with “regional
124 specialties” specifically covering recipes for regional model configurations (Barnes et al.,
125 2024).

126 The intended mode of use is also explicit. The repository is designed around Jupyter-
127 based analysis on the Australian National Computational Infrastructure, with guidance
128 on running notebooks through ARE/JupyterLab sessions and on accessing shared data
129 holdings. This makes the Cookbook more than a static collection of examples: it is
130 operational documentation for a real analysis environment. At the same time, the
131 notebooks demonstrate broadly transferable patterns for working with labelled geophysical
132 data, so individual lessons can be reused or adapted outside that environment.

133 The project also supports community authorship. Contributors are encouraged to submit
134 new notebooks through pull requests, document their methods clearly, and generalise
135 workflows so that they remain useful to others. This is particularly valuable as it is
136 relatively rare for the computer code underlying scientific analyses to be peer reviewed
137 with the same care as the manuscript itself. The COSIMA community provides several
138 examples of the recipe ecosystem being extended into full research projects built on
139 peer-authored software repositories on Github, where analysis workflows, methods, and
140 implementation choices are exposed to community scrutiny and reuse. In that sense, the
141 Cookbook functions both as a learning module for end users and as a framework for
142 teaching reproducible scientific communication through notebook design and peer review.
143 The easy access to data and analysis provided by the COSIMA Cookbook showcases best
144 practices to the broader oceanographic community and has enabled rapid community
145 adoption of the ACCESS ocean and sea-ice model configurations internationally, facilitating
146 to date well over 100 peer-reviewed papers and more than 20 PhD projects to completion
147 with several other underway.

148 Author Order

149 The first two authors contributed equally; authors from the third position onward are
150 listed alphabetically by surname.

Acknowledgements

This work was supported by computational resources provided by the Australian Government through the National Computational Infrastructure (NCI) under the National Computational Merit Allocation Scheme and the Australian National University Allocation Scheme. We thank the vibrant communities of the Consortium for Ocean–Sea Ice Modelling in Australia (COSIMA; cosima.org.au) and Australia’s climate simulator (ACCESS-NRI; access-nri.org.au) for making the ACCESS-OM2 and ACCESS-OM3 outputs and analysis tools available through the NCI. We further acknowledge funding from the Australian Research Council under the Centre of Excellence for the Weather of the 21st Century CE230100012, the Linkage Infrastructure, Equipment and Facilities LP160100073 and LP200100406, and the Discovery Project DP240101274.

References

- Abernathy, R. P., Busecke, J. J. M., Smith, T. A., Deauna, J. D., Banihirwe, A., Nicholas, T., Fernandes, F., James, B., Dussin, R., Cherian, D. A., Caneill, R., Sinha, A., Uieda, L., Rath, W., Balwada, D., Constantinou, N. C., Ponte, A., Zhou, Y., Uchida, T., & Thielen, J. (2022). *xgcm* (Version v0.8.1). Zenodo. <https://doi.org/10.5281/zenodo.7348619>
- Barnes, A. J., Constantinou, N. C., Gibson, A. H., Kiss, A. E., Chapman, C., Reilly, J., Bhagtani, D., & Yang, L. (2024). regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, 9(100), 6857. <https://doi.org/10.21105/joss.06857>
- Constantinou, N. C., Hogg, A. McC., Gibson, A., Steketee, A., Beucher, R., Proft, M., Heerdegen, A., Neme, J., Holmes, R. M., Morrison, A., Fierro-Arcos, D., Kiss, A. E., Oliveira, M., Yung, C., Doddridge, E., Huneke, W., Bhagtani, D., Ong, E. Q. Y., Munroe, J., ... Narayanan, A. (2026). *COSIMA Cookbook: A cookbook of recipes for analysing ocean and sea ice model output*. <https://doi.org/10.5281/zenodo.14353852>
- Dask Development Team. (2016). *Dask: Library for dynamic task scheduling*. <http://dask.pydata.org>
- Delandmeter, P., & Seville, E. van. (2019). The parcels v2.0 lagrangian framework: New field interpolation schemes. *Geoscientific Model Development*, 12(8), 3571–3584. <https://doi.org/10.5194/gmd-12-3571-2019>
- Elson, P., Andrade, E. S. de, Lucas, G., May, R., Hattersley, R., Campbell, E., Comer, R., Dawson, A., Little, B., Raynaud, S., scm72, Snow, A. D., Igolston, Blay, B., Killick, P., Ibdreyer, Peglar, P., Wilson, N., Andrew, ... Kirkham, D. (2024). *SciTools/cartopy: v0.24.1* (Version v0.24.1). Zenodo. <https://doi.org/10.5281/zenodo.13905945>
- Hoyer, S., & Hamman, J. J. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal of Open Research Software*, 5(1), 10. <https://doi.org/10.5334/jors.148>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Intake developers. (2026). *Intake: A lightweight package for finding, investigating, loading and disseminating data*. <https://intake.readthedocs.io/en/reader/>
- Kiss, A. E., Woodward, M. J., Hollings, T., Mu, M., Fiedler, R., Steven, A. D. L., Lenton, A., Matear, R., Chamberlain, M. A., Oke, P. R., Alves, O., Griffies, S. M., Hill, C., Kovach, A., Marsland, S. J., Fiedler, R., Waters, J., Yang, C.-H., Zhou, X., ... Hogg, A. McC. (2020). ACCESS-OM2 v1.0: A global ocean-sea ice model at three resolutions. *Geoscientific Model Development*, 13(2), 401–442. <https://doi.org/10.5194/gmd-13-401-2020>

197 401-2020

- 198 Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J.,
199 Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S.,
200 & Willing, C. (2016). Jupyter notebooks—a publishing format for reproducible
201 computational workflows. In F. Loizides & B. Schmidt (Eds.), *Positioning and power
202 in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. <https://doi.org/10.3233/978-1-61499-649-1-87>
203
- 204 Komiya, T., Brandl, G., Turner, A., Bakosi, J.-F., Shimizukawa, T., Andersen, J. L.,
205 Neuhäuser, D., Finucane, S., Lehmann, R., Mason, J., Kampik, T., Addison, J., Magin,
206 J., Dufresne, J., Waltman, J., Eades, D., Cano Rodríguez, J. L., Ronacher, A., Geier,
207 M., ... Freitag, F. (2025). *sphinx-doc/sphinx: Sphinx 9.1.0* (Version v9.1.0). Zenodo.
208 <https://doi.org/10.5281/zenodo.18109073>
- 209 Rocklin, M. (2015). Dask: Parallel computation with blocked algorithms and task
210 scheduling. In K. Huff & J. Bergstra (Eds.), *Proceedings of the 14th python in science
211 conference* (pp. 130–136). <https://doi.org/10.25080/Majora-7b98e3ed-013>
- 212 Zhuang, J., Dussin, R., Bourgault, P., Huard, D., Banihirwe, A., Raynaud, S., Malevich,
213 B., Schupfner, M., Gauthier, C., Filipe, S., Almansí, M., Jüling, A., Scott,
214 R., RondeauG, Rasp, S., Smith, T. J., Meyer, A. G., Mares, B., ... Li, X. (2025).
215 *pangeo-data/xESMF: v0.9.2*. <https://doi.org/10.5281/zenodo.4294774>